# CompositionMeasure for PureDP

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `CompositionMeasure` for `PureDP` in `mod.rs` at commit f5bb719 (outdated[1]).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

Types matching pseudocode.

#### Caller-verified

None

### Pseudocode

```
1  class CompositionMeasure(PureDP):
2      def composability(  #
3          self, adaptivity: Adaptivity
4      ) -> Composability:
5          return Composability.Concurrent
6
7      def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
8          d_out = 0.0
9          for d_mid in d_mids:
10             d_out = d_out.inf_add(d_mid)
11         return d_out
```

### Postcondition

**Theorem 1.1.** `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` adaptivity and `out`-composability. Otherwise returns an error.

*Proof.* By the postcondition of `InfAdd` we have that $\sum_i \texttt{d_mids}_i \leq \texttt{compose(d_mids)}$.

| Adaptivity | Sequential | Concurrent |
|---|---|---|
| Non-Adaptive | Theorem 1[DMNS06] | Theorem 1.8[VW21] |
| Adaptive | Theorem 1[DMNS06] | Theorem 1.8[VW21] |
| Fully-Adaptive | None | None |

This table is reflected in the implementation of `composability` on line 2.

[1] See new changes with `git diff f5bb719..dde8678d rust/src/combinators/sequential_composition/mod.rs`

□

## References


[DMNS06] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In Shai Halevi and Tal Rabin, editors, *Theory of Cryptography*, pages 265–284, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.

[VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.